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Current-limiting control strategy for single-loop droop-controlled grid-forming inverters

Abstract

This document describes systems and techniques for a current-limiting control strategy for single-loop droop-controlled grid-forming inverters. In aspects, a hysteresis module is configured to compare an output current detected across one or more transistors in an inverter controlled by the single-loop droop converter with a specified maximum current and to generate an overcurrent signal. The overcurrent signal presents a fault signal responsive to the output current exceeding the specified maximum current. A logic array is configured to logically combine gate control signals generated by the single-loop droop controller to selectively direct the one or more transistors to allow the output current to flow therethrough with the overcurrent signal to present modified gate control signals to the one or more transistors. The logic array is configured to replace one or more of the gate control signals in the modified gate control signals with a gate disable signal responsive to the overcurrent signal presenting the fault signal.

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Background/Summary

INCORPORATION BY REFERENCE (1) This application claims the benefit of U.S. Provisional Patent Application No. 63/404,060, filed Sep. 6, 2022, and titled “A CURRENT LIMITING CONTROL STRATEGY FOR SINGLE-LOOP DROOP-CONTROLLED GRID-FORMING INVERTERS UNDER BALANCED AND UNBALANCED FAULTS” (the “Provisional Application”). This application incorporates by reference the entirety of the Provisional Application.

BACKGROUND

(1) Grid-forming inverters are increasingly used to improve the stability of a power grid by enabling renewable power sources to supply power to the power grid to supplement conventional generating sources. Most renewable sources, such as solar power systems and wind power systems, typically provide sources of direct current (DC) power, either directly or via storage batteries in which the DC power generated is stored for later use. Although power grids operate using alternating current (AC) power, grid-forming inverters convert DC power sources to AC power for supply to the power grid.

(2) To a power grid, a grid-forming inverter fundamentally behaves as a voltage source behind an impedance. A potential disadvantage of the grid-forming inverter is that transistors or other components of grid-forming inverters may be damaged during overcurrent episodes, such as occurrence of short circuits, if protective measures are not used.

(3) Droop control is a widely used technique for managing current flow in a grid-forming inverter to one of most mature grid-forming control strategies. Two common forms of droop-control systems are multi-loop droop-controlled systems and single-loop droop-controlled systems. A multi-loop droop-controlled system typically controls the magnitude and frequency of an inverter filter capacitor voltage according to the droop laws using cascaded inner voltage and current loops to achieve fast control of the filter capacitor voltage. A single-loop droop-controlled system directly controls magnitude and frequency of a modulation waveform according to the droop laws without using a cascaded inner current loop. Multi-loop droop-controlled systems often focus on modifying a current reference of the cascaded inner current loop of multi-loop droop-controlled systems to limit overcurrent episodes, but these methods are not applicable to single-loop droop-controlled structures that do not use a cascaded inner current loop. A single-loop droop-controlled system—as the name may imply—is simpler in structure than a multi-loop droop-controlled system, but a single-loop droop-controlled system may be more vulnerable to overcurrent episodes than multi-loop droop-controlled systems.

SUMMARY

(4) This document describes systems and techniques for a current-limiting control strategy for single-loop droop-controlled grid-forming inverters. In aspects, a hysteresis module is configured to compare an output current detected across one or more transistors in an inverter controlled by the single-loop droop converter with a specified maximum current and to generate an overcurrent signal. The overcurrent signal presents a fault signal responsive to the output current exceeding the specified maximum current. A logic array is configured to logically combine gate control signals generated by the single-loop droop controller to selectively direct the one or more transistors to allow the output current to flow therethrough with the overcurrent signal to present modified gate control signals to the one or more transistors. The logic array is configured to replace one or more of the gate control signals in the modified gate control signals with a gate disable signal responsive to the overcurrent signal presenting the fault signal.

(5) This Summary introduces simplified concepts related to systems and techniques for a current-limiting control strategy for single-loop droop-controlled grid-forming inverters, as further described in the Detailed Description and Drawings. This Summary is not intended to identify essential features of the claimed subject matter, nor is it intended for use in determining the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The details of one or more aspects of systems and techniques for a current-limiting control strategy for single-loop droop-controlled grid-forming inverters are described in this document

with reference to the following drawings, wherein the same numbers are used throughout the drawings to reference like features and components:

- (2) FIG. 1 is a schematic diagram of a single-loop droop-controlled grid forming inverter using an example current-limiting control system;
- (3) FIG. 2 is an enlarged schematic diagram of an array of transistors configured as an inverter included in the single-loop droop-controlled grid forming inverter of FIG. 1;
- (4) FIG. 3 is a schematic diagram of the current-limiting control system of FIG. 2 including a hysteresis loop and a logic array;
- (5) FIG. 4A is a schematic diagram of the inverter of FIG. 2 in an overcurrent situation without engagement of the current-limiting control system;
- (6) FIG. 4B is a partial schematic diagram of the inverter of FIG. 4A depicting an effect of the overcurrent situation depicted in FIG. 4A;
- (7) FIG. 5A is a schematic diagram of the inverter of FIG. 2 in an overcurrent situation with the current-limiting control system engaged to respond to the overcurrent situation;
- (8) FIG. 5B is a partial schematic diagram of the inverter of FIG. 5A depicting the response of the current-limiting control system to the overcurrent situation;
- (9) FIG. 6 is a composite graph of an output current in an overcurrent situation in the single-loop droop-controlled grid forming inverter with and without response of the current-limiting control system;
- (10) FIG. 7 is a graph of the output current in the overcurrent situation depicted in FIG. 6 depicting an effect of the current-limiting control system redirecting the output current; and
- (11) FIG. 8 is a flow diagram of an example method of limiting current across one or more transistors in a grid-forming inverter.

DETAILED DESCRIPTION

(12) Overview

(13) This document describes systems and techniques for a current-limiting control strategy for single-loop droop-controlled grid-forming inverters. In aspects, a hysteresis module is configured to compare an output current detected across one or more transistors in an inverter controlled by the single-loop droop converter with a specified maximum current and to generate an overcurrent signal. The overcurrent signal presents a fault signal responsive to the output current exceeding the specified maximum current. A logic array is configured to logically combine gate control signals generated by the single-loop droop controller to selectively direct the one or more transistors to allow the output current to flow therethrough with the overcurrent signal to present modified gate control signals to the one or more transistors. The logic array is configured to replace one or more of the gate control signals in the modified gate control signals with a gate disable signal responsive to the overcurrent signal presenting the fault signal.

Example Systems

(14) FIG. 1 illustrates a single-loop droop-controlled grid-forming inverter **100**, such as may be used to couple a DC power source (V_{DC}) **102** to a power grid (not shown) at an output **104**. The single-loop droop-controlled grid-forming inverter **100** includes an example current-limiting control system **106**. The current-limiting control system **106** includes a hysteresis loop **108** and a logic array **110**. The hysteresis loop **108** and the logic array **110** are configured to monitor output currents *i.sub.a* **112**, *i.sub.b* **114**, and *i.sub.c* **116** generated by an inverter **118** to detect and respond to overcurrent situations. An overcurrent situation is a situation in which one or more of the output currents *i.sub.a* **112**, *i.sub.b* **114**, and *i.sub.c* **116** exceed a maximum current threshold at or beyond which the output current may damage transistors or other components included in the inverter **118**. Such conditions may arise, for example, during a short circuit or as a result of a power surge occurring at the DC power source **102** which results in an excessive current or “fault current” flowing in the inverter **118**. The current-limiting control system **106** is configured to detect an overcurrent situation by responding to one or more the output currents *i.sub.a* **112**, *i.sub.b* **114**, and

i.sub.c **116** exceeding the maximum current threshold and redirecting fault currents before they can cause damage to the components of the inverter **118**, as further described below.

(15) The inverter **118**, for purposes of this example, is a three-phase bridge inverter of a type commonly used to convert DC power into sinusoidal-wave AC power. The inverter **118** includes three branches **120**, **122**, and **124**, each of which includes an output node **126**, **128**, **130** at which the output currents i.sub.a **112**, i.sub.b **114**, and i.sub.c **116**, respectively, are presented.

(16) Each of the three branches **120**, **122**, and **124** includes a pair of transistors coupled to opposing sides of a DC bus **132**. A first branch **120** includes a high-side transistor T.sub.1 **138** having its collector coupled to the positive side **134** of the DC bus **132** and an emitter coupled to a collector of a low-side transistor T.sub.4 **140** of which an emitter is coupled to the negative side **136** of the DC bus **132**. The first output node **126** lies at the interconnection of the emitter of the high-side transistor T.sub.1 **138** and the collector of the low-side transistor T.sub.4 **140**. A second branch **122** includes a high-side transistor T.sub.3 **142** having its collector coupled to the positive side **134** of the DC bus **132** and an emitter coupled to a collector of a low-side transistor T.sub.6 **144** of which an emitter is coupled to the negative side **136** of the DC bus **132**. The second output node **128** lies at the interconnection of the emitter of the high-side transistor T.sub.3 **142** and the collector of the low-side transistor T.sub.6 **144**. A third branch **124** includes a high-side transistor T.sub.5 **146** having its collector coupled to the positive side **134** of the DC bus **132** and an emitter coupled to a collector of a low-side transistor T.sub.2 **148** of which an emitter is coupled to the negative side **136** of the DC bus **132**. The third output node **130** lies at the interconnection of the emitter of the high-side transistor T.sub.5 **146** and the collector of the low-side transistor T.sub.2 **148**. In various aspects, each of the transistors **138**, **140**, **142**, **144**, **146**, and **148** includes an insulated-gate bipolar transistor (IGBT) having a high capacity for carrying bipolar current between its collector and emitter.

(17) As further described with reference to FIG. 2, gates of the transistors **138**, **140**, **142**, **144**, **146**, and **148** of the different branches **120**, **122**, and **124** are selectively activated to generate the output currents i.sub.a **112**, i.sub.b **114**, and i.sub.c **116** at the output nodes **126**, **128**, and **130**, respectively, that ultimately generate AC power at voltage outputs v.sub.ga **150**, v.sub.gb **152**, and v.sub.gc **154** at the output **104**. Depending upon gate control signals (further described below) applied to the gates of the transistors **138**, **140**, **142**, **144**, **146**, and **148**, the output currents i.sub.a **112**, i.sub.b **114**, and i.sub.c **116** flow across one or more of the transistors **138**, **140**, **142**, **144**, **146**, and **148**. An object of the current-limiting control system **106** is to detect when one or more of the output currents i.sub.a **112**, i.sub.b **114**, and i.sub.c **116** presents a fault current that could damage one or more of the transistors **138**, **140**, **142**, **144**, **146**, and **148** and then to apply signals to gates of the corresponding one or more of the transistors **138**, **140**, **142**, **144**, **146**, and **148** to prevent the fault current from damaging the one or more of the transistors **138**, **140**, **142**, **144**, **146**, and **148**.

(18) As is understood by those skilled in droop control, a single-loop droop-controlled grid-forming inverter **100** employs a sensor **158** to monitor the output **104**. An instantaneous active power (P), reactive power (Q), and voltage magnitude (V.sub.mag) calculation and low-pass filter **160** receives data from the sensor **158**. An output of the P, Q, and V.sub.mag calculation and low-pass filter **160** are presented to a single-loop droop-control and overload mitigation controller **162**. In turn, an output of the single-loop droop-control and overload mitigation controller **162** is presented to a pulse-width modulator **164**. The pulse-width modulator **164** generates a series of gate control signals **166** (designated g.sub.1, g.sub.2, g.sub.3, g.sub.4, g.sub.5, and g.sub.6 in FIG. 1) that are configured to control the transistors **138**, **140**, **142**, **144**, **146**, and **148** in the inverter **118** to generate the output currents i.sub.a **112**, i.sub.b **114**, and i.sub.c **116**. The pulse-width modulator **164** may be adapted to generate sinusoidal-wave AC power based on controlling the timing and magnitude of DC pulses presented at the output current nodes **126**, **128**, and **130**. In a conventional single-loop droop-controlled grid-forming inverter (not shown), the gate control signals **166** would be presented directly to the transistors **138**, **140**, **142**, **144**, **146**, and **148** in the inverter **118**.

(19) In various aspects, instead of presenting the gate control signals **166** directly to the transistors **138, 140, 142, 144, 146, and 148** in the inverter **118**, the gate control signals **166** are presented to the current-limiting control system **106**. The current-limiting control system **106** is coupled with a current sensor **168** configured to monitor the output currents $i_{\text{sub.a}}$ **112**, $i_{\text{sub.b}}$ **114**, and $i_{\text{sub.c}}$ **116** flowing between the output nodes **126, 128, 130**. As described in detail below, the hysteresis loop **108** compares the output currents $i_{\text{sub.a}}$ **112**, $i_{\text{sub.b}}$ **114**, and $i_{\text{sub.c}}$ **116** with a specified maximum current **170** and generates corresponding overcurrent signals $b_{\text{sub.a}}$ **172**, $b_{\text{sub.b}}$ **174**, and $b_{\text{sub.c}}$ **176**. When the hysteresis loop **108** detects that one or more of the output currents $i_{\text{sub.a}}$ **112**, $i_{\text{sub.b}}$ **114**, and $i_{\text{sub.c}}$ **116** exceed the specified maximum current **170**, the hysteresis loop **108** causes one or more of the overcurrent signals $b_{\text{sub.a}}$ **172**, $b_{\text{sub.b}}$ **174**, and $b_{\text{sub.c}}$ **176** to present a fault signal.

(20) In various aspects, the logic array **110** receives the gate control signals **166** from the pulse-width modulator **164** and the overcurrent signals $b_{\text{sub.a}}$ **172**, $b_{\text{sub.b}}$ **174**, and $b_{\text{sub.c}}$ **176** and to logically combine corresponding signals. Responsive to one or more of the overcurrent signals $b_{\text{sub.a}}$ **172**, $b_{\text{sub.b}}$ **174**, and $b_{\text{sub.c}}$ **176** presenting the fault signal, the logic array **110** replaces one or more of the gate control signals **166** presented by the pulse-width modulator **164** in the modified gate control signals **178** (designated $g_{\text{sub.1'}}$, $g_{\text{sub.2'}}$, $g_{\text{sub.3'}}$, $g_{\text{sub.4'}}$, $g_{\text{sub.5'}}$, and $g_{\text{sub.6'}}$ in FIG. 1) with a gate disable signal. Presentation of the gate disable signal prevents the fault current detected at one or more of the output currents $i_{\text{sub.a}}$ **112**, $i_{\text{sub.b}}$ **114**, and $i_{\text{sub.c}}$ **116** from flowing through one or more of the transistors **138, 140, 142, 144, 146, and 148**. Instead of the fault current flowing through one or more of the transistors **138, 140, 142, 144, 146, and 148**, the fault current is redirected, preventing the fault current from damaging one or more of the transistors **138, 140, 142, 144, 146, and 148**.

(21) Referring to FIG. 2, each of the transistors **138, 140, 142, 144, 146, and 148** of the inverter **118** has a diode **200, 202, 204, 206, 208, and 210**, coupled in a blocking configuration between a collector **220, 222, 224, 226, 228, and 230** and an emitter **240, 242, 244, 246, 248, and 250** of each of the transistors **138, 140, 142, 144, 146, and 148**, respectively. In this configuration, the anode of each of the diodes **200, 202, 204, 206, 208, and 210** is coupled to the emitter and its cathode is coupled to the collector of its respective transistor **138, 140, 142, 144, 146, and 148**, respectively. As a result, each of the diodes **200, 202, 204, 206, 208, and 210** opposes a flow of current around the respective transistors **138, 140, 142, 144, 146, and 148** from the collector **220, 222, 224, 226, 228, 230** to the emitter **240, 242, 244, 246, 248, and 250**. However, the diodes **200, 202, 204, 206, 208, 210** permit a flow of current around the respective transistors **138, 140, 142, 144, 146, and 148** from the emitter **240, 242, 244, 246, 248, and 250** to the collector **220, 222, 224, 226, 228, and 230**. Thus, as described further below, an excessive current detected across one or more of the transistors **138, 140, 142, 144, 146, and 148** may be redirected to the DC bus **132** where it may be discharged rather than be permitted to flow across one or more of the transistors **138, 140, 142, 144, 146, and 148**.

(22) For example, when a gate $g_{\text{sub.1}}$ **252** of the transistor $T_{\text{sub.1}}$ **138** is enabled, a first current $i_{\text{sub.1}}$ **254** applied at the collector **220** of the transistor $T_{\text{sub.1}}$ **138** at the positive side **134** of the DC bus **132** is permitted to flow through the transistor $T_{\text{sub.1}}$ **138** to the emitter **240** of the transistor $T_{\text{sub.1}}$ **138**. The current $i_{\text{sub.1}}$ **254** is presented at the first output node **126** and also may flow across the transistor $T_{\text{sub.4}}$ **140** (if a gate $g_{\text{sub.4}}$ **256** is enabled). On the other hand, when the gate $g_{\text{sub.1}}$ **252** is disabled, the first current $i_{\text{sub.1}}$ **254** applied at the collector **220** of the transistor $T_{\text{sub.1}}$ **138** at the positive side **134** of the DC bus **132** is prevented from flowing through the transistor $T_{\text{sub.1}}$ **138** to the emitter **240** of the transistor $T_{\text{sub.1}}$ **138**. Similarly, when the gate $g_{\text{sub.4}}$ **256** is disabled, a current detected at the output node **126** applied at the collector **230** of the transistor $T_{\text{sub.4}}$ **140** is prevented from flowing through the transistor $T_{\text{sub.4}}$ **140** to the emitter **242** of the transistor $T_{\text{sub.4}}$ **140**. Controlling the flow of current through the transistor $T_{\text{sub.1}}$ **138** and/or the transistor $T_{\text{sub.4}}$ **140** (as well as through the transistors **142, 144, 146, and**

148) is how the inverter **118**, under control of the gate control signals **166** generated by the pulse-width modulator **164** (FIG. 1), generates sinusoidal AC current at the output **104**.

(23) Although an IGBT transistor like T.sub.1 **138** is rated to accommodate a large bipolar current flowing between the collector **220** and the emitter **240**, the transistor T.sub.1 **138** may nonetheless be damaged if the current i.sub.1 **254** applied at a collector **220** exceeds a specified maximum current for which the transistor T.sub.1 **138** is rated. Accordingly, by disabling the gate g.sub.1 **212**, the current i.sub.1 **254** is prevented from flowing through the transistor T.sub.1 **138**, protecting the transistor T.sub.1 **138** from damage. Similarly, by disabling the gate g.sub.4 **256**, a current at the output node **126** is prevented from flowing through the transistor T.sub.4 **140**, protecting the transistor T.sub.4 **140** from damage. Thus, in response to a current applied across one or more of the transistors **138**, **140**, **142**, **144**, **146**, and **148**, the current-limiting control system **106** disables gates of the respective transistors **138**, **140**, **142**, **144**, **146**, and **148** to prevent them from being damaged by a fault current in excess of a specified maximum current **170**.

(24) In various aspects, when a fault current is detected by the current-limiting control system **106**, such as if the current i.sub.1 **254** or another current detected at one of the output nodes **126**, **128**, and/or **130** is determined to be in excess of the specified maximum current **170**, the current may be redirected to the DC bus **132**. For example, when a gate g.sub.4 **256** of the transistor T.sub.4 **140** is disabled, a current detected at the output node **126** cannot flow through the transistor T.sub.4 **140**, but it may flow across the diode **200**, as a diode current d.sub.1 **258** back to the positive side **134** of the DC bus **132** where the current d.sub.1 **258** may dissipate. The fault current may flow “backward” through the DC bus **132** where it may flow from the negative side **136** of the DC bus through the diodes **200**, **202**, **204**, **206**, **208**, and/or **210** one or more branches **120**, **122**, and **124** of the inverter **118** until the fault current is dissipated.

(25) Thus, in response to detecting a fault current at one or more of the output nodes **126**, **128**, and/or **130**, the current-limiting control system **106** thus protects the transistors **138**, **140**, **142**, **144**, **146**, and **148** by disabling gates of the respective transistors **138**, **140**, **142**, **144**, **146**, and **148** to prevent the fault current from flowing through the respective transistors **138**, **140**, **142**, **144**, **146**, and **148**. The current-limiting control system **106** disables the gates of the affected transistors by replacing the gate control signals **166** generated by the pulse-width modulator **164** with modified gate control signals **178**, as further described below with reference to FIG. 3. The fault current is discharged to the DC bus **132** through one or more of the diodes **200**, **202**, **204**, **206**, **208**, and/or **210**. When gates of the respective transistors **138**, **140**, **142**, **144**, **146**, and/or **148** are reenabled, if the fault current is again detected at one or more of the output nodes **126**, **128**, and/or **130**, gates of the respective transistors **138**, **140**, **142**, **144**, **146**, and/or **148** are again disabled. The sequence repeats until the fault current is no longer detected at one or more of the output nodes **126**, **128**, and/or **130**.

(26) FIG. 3 details an example configuration of the hysteresis loop **108** and the logic array **110** of the current-limiting control system **106**. The hysteresis loop **108** receives current magnitudes **300**, **302**, and **304** of the currents i.sub.a **112**, i.sub.b **114**, and i.sub.c **116**, respectively, and the specified maximum current **170**. The respective current magnitudes **300**, **302**, and **304** and the specified maximum current are supplied to comparators **310**, **320**, and **330**. Specifically, the specified maximum current **170** is presented to positive inputs **312**, **322**, and **324** of the comparators **310**, **320**, and **330**. Current magnitudes **300**, **302**, and **304** of the currents i.sub.a **112**, i.sub.b **114**, and i.sub.c **116** are presented to negative inputs **314**, **324**, and **334** of the comparators **310**, **320**, and **330**. As long as the specified maximum current **170** is greater than the respective current magnitudes **300**, **302**, and **304**, outputs **316**, **326**, and **336** of the respective comparators **310**, **320**, and **330** will be a logical 1 or a high output. By contrast, when the respective current magnitudes **300**, **302**, and **304** are greater than the specified maximum current **170** is greater than, the outputs **316**, **326**, and **336** of the respective comparators **310**, **320**, and **330** will be a logical 0 or a low output. In various aspects, the outputs **316**, **326**, and **336** are presented to buffers **318**, **328**, and **338**,

respectively. Outputs of the buffers b.sub.a 172, b.sub.b 174, and b.sub.c 176 are termed “overcurrent signals” in that the outputs of the buffers b.sub.a 172, b.sub.b 174, and b.sub.c 176 will indicate when one of more of the currents i.sub.a 112, i.sub.b 114, and i.sub.c 116 includes a fault current in excess of the specified maximum current 170.

(27) The overcurrent signals b.sub.a 172, b.sub.b 174, and b.sub.c 176 are presented to the logic array 110 along with the gate control signals 166 generated by the pulse-width modulator 164 (FIG. 1), g.sub.1 340, g.sub.4 342, g.sub.3 344, g.sub.6 346, g.sub.5 348, and g.sub.2 350. The logic array 110 includes three pairs of logic elements 360, 362, and 364. In this example, the pairs of logic elements 360, 362, and 364 include AND gates 370, 372, 374, 376, 378, and 380. At each of the AND gates 370, 372, 374, 376, 378, and 380, one of the overcurrent signals b.sub.a 172, b.sub.b 174, and b.sub.c 176 is logically combined with one of the gate control signals g.sub.1 340, g.sub.4 342, g.sub.3 344, g.sub.6 346, g.sub.5 348, and g.sub.2 350.

(28) Specifically, at a first pair of logic elements 360, the overcurrent signal b.sub.a 172 is logically combined at the AND gate 370 with the gate control signal g.sub.1 340 to generate a modified gate control signal g.sub.1' 382 and the overcurrent signal b a 172 is logically combined at the AND gate 372 with the gate control signal g.sub.4 342 to generate a modified gate control signal g.sub.4' 384. The gate control signal g.sub.1 342 may be paired in the pair of logic elements 360 with the gate control signal g.sub.4 342 to correspond to pairing5 of transistors T.sub.1 138 and T.sub.4 140 in the first branch 120 of the inverter 118 (FIGS. 1 and 2). Similarly, in a second pair of logic elements 362, the overcurrent signal b.sub.b 174 is logically combined at the AND gate 374 with the gate control signal g.sub.3 344 to generate a modified gate control signal g.sub.3' 386 and the overcurrent signal b.sub.b 174 is logically combined at the AND gate 376 with the gate control signal g.sub.6 346 to generate a modified gate control signal g.sub.6' 388. In a third pair of logic elements 364, the overcurrent signal b.sub.c 176 is logically combined at the AND gate 378 with the gate control signal g.sub.5 348 to generate a modified gate control signal g.sub.5' 390 and the overcurrent signal b.sub.c 176 is logically combined at the AND gate 380 with the gate control signal g.sub.2 350 to generate a modified gate control signal g.sub.2' 392.

(29) The logic array 110 thus is responsive to the hysteresis loop 108 to disable gates of transistors with a gate disable signal where a fault current has been detected across the respective transistor. For example, when i.sub.a 112 is identified by the comparator 310 of the hysteresis loop 108 as a fault current, the overcurrent signal b a 172 includes a fault signal, which is a logical 0 or low signal. When combined with the gate control signals g.sub.1 340 and g.sub.4 342 at the AND gates 370 and 372, the modified gate control signals g.sub.1' 382 and g.sub.4' 384 replace the gate control signals g.sub.1 340 and g.sub.4 342 with logical 0 or low gate disable signals to disable gates g.sub.1 252 and g.sub.4 256 of the transistors T.sub.1 138 and T.sub.4 140, respectively. Thus, regardless of the state of the gate control signals g.sub.1 340 and g.sub.4 342 presented by the pulse-width modulator 164, the current-limiting control system 106 will the replace gate control signals g.sub.1 340 and g.sub.4 342 with gate disable signals to disable the gates g.sub.1 252 and g.sub.4 256 of the transistors T.sub.1 138 and T.sub.4 140, respectively. The current-limiting control system 106 thus may prevent the fault current from flowing through—and potentially, damaging—the transistors T.sub.1 138 and T.sub.4 140 or other transistors 142, 144, 146, and 148 of the inverter 118.

(30) FIGS. 4A-5B illustrate an example of how implementations of the current-limiting control system 106 may redirect a fault current 400 detected at output node 126 to avoid damage to transistors T.sub.1 138 and T.sub.4 140. Fault currents detected at output nodes 126 and 128 may b.sub.e redirected to avoid damage to transistors 142, 144, 146, and 148 in a similar way. In the example of FIGS. 4A-5B, the fault current 400 is a result of a short circuit 402 between output nodes 126 and 128, although this is just one example of a cause of a fault current. FIGS. 4A and 4B illustrate a case in which the current-limiting control system 106 is not engaged with the inverter 118 to respond to the fault current 400. FIGS. 5A and 5B illustrate a case in which the current-

limiting control system **106** is engaged with the inverter **118** to respond to the fault current **400**.

(31) Referring to FIG. **4A**, when the short circuit **402** causes the fault current **400** to b.sub.e presented, the gate control signals **166** presented by the pulse-width modulator **164** (not shown in FIGS. **4A-5B**) include an active gate control signal g.sub.1=1 **404** to enable the gate g.sub.1 **252** of transistor T.sub.1 **138** to conduct current between the collector **220** and the emitter **240** and an active gate control signal g.sub.6=1 **406** to enable a gate g.sub.6 **404 52** of transistor T.sub.6 **144** to conduct current between the collector **226** and the emitter **246**. As a result, referring to FIG. **4B**, the fault current **400** flows through the short circuit **402** through the transistors T.sub.1 **138** and T.sub.6 **144**. The fault current **400**, being in excess of the specified maximum current **170** for which the transistors T.sub.1 **138** and T.sub.6 **144** are rated, may potentially damage the transistors T.sub.1 **138** and T.sub.6 **144** and damage the inverter **118**.

(32) However, the current-limiting control system **106** may prevent the short circuit **402** from causing damage to the transistors T.sub.1 **138** and T.sub.6 **144**. Referring to FIG. **5A**, responsive to detecting the fault current **400** across the transistors T.sub.1 **138** and T.sub.6 **144**, as described with reference to FIG. **3**, the current-limiting control system **106** replaces the active gate control signals g.sub.1=1 **404** and g.sub.6=1 **406** of the transistors T.sub.1 **138** and T.sub.6 **144** with gate disable signals g.sub.1=0 **504** and g.sub.6=0 **506**, respectively. As a result, the transistors T.sub.1 **138** and T.sub.6 **144** no longer conduct the fault current **400** and the fault current **400** flows through the diodes **200** and **206** across the collectors **220** and **226** and emitters **240** and **246** of the transistors T.sub.1 **138** and T.sub.6 **144**, respectively. The redirected fault current **400** flows back onto the DC bus **132** where it is dissipated instead of flowing through— and potentially damaging—the transistors T.sub.1 **138** and T.sub.6 **144**.

(33) FIG. **6** is a graph **600** comparing a plot of a current **602** detected across a plurality of transistors in an overcurrent situation with a plot of a modified current **604** constrained by the current-limiting control system **106**. At a time the fault current **602** is detected, t.sub.f **606**, the current **602** exceeds a specified maximum current value **608** at numerous points and, thus, presents a fault current. By contrast, the modified current **604** is contained within a range bounded by the specified maximum current **608** by the current-limiting control system **106**. As previously described, when the current-limiting control system **106** detects a current that exceeds the specified maximum current value **608**, the current-limiting control system **106** disables one or more of the transistors to cut off the flow of current through the transistors and redirects the excessive, fault current back across the DC bus **132** (see FIGS. **5A** and **5B**) until the fault current is dissipated. Thus, the current-limiting control system **106** prevents the current **602** from flowing through the transistors and other components when the current **602** exceeds a specified maximum current value **608** and, thus, may cause damage to components through which the current passes.

(34) FIG. **7** is a graph **700** of an enlarged region **610** (FIG. **6**) of the plot of current **602**. As previously described, upon detecting a current in excess of the specified maximum current value **608**, the current-limiting control system **106** disables gates of one or more transistors through which the fault current would flow and redirects the fault current back to the DC bus. Thus, a plot of current flowing through the one or more affected transistors **702** is cut off at or around a level of the specified maximum current value **608**. Then, as described with reference to FIGS. **5A** and **5B**, a redirected current **704**, redirected through one or more diodes such as diodes **200** and **206**, may circulate back through the DC bus **132**. If the redirected current **704** results in an output current across one or more of the transistors that exceeds the specified maximum current value **608**, that current also will be constrained by disabling the transistors through which the redirected current **704** flows until the redirected current dissipates. Thus, by disabling transistors across which a fault current is applied to redirect the fault current, then by continuing to rapidly enable and disable other transistors in response to whether the fault current is detected across the respective transistors, the current-limiting control system **106** repeatedly redirects the fault current to the DC bus until the fault current is dissipated.

Example Method

(35) FIG. 8 is a flow diagram of an example method **800** for limiting current across one or more transistors in a grid-forming inverter controlled by a single-loop droop controller as previously described with reference to FIGS. 1-7. At a block **802**, a plurality of gate control signals are generated, the gate control signals being applicable to a plurality of transistors in the grid-forming inverter to cause the grid-forming inverter to generate a plurality of output currents combinable to form an alternating current. As described with reference to FIG. 1, a pulse-width modulator **164** may be used to generate gate control signals configured to enable the plurality of transistors to generate DC currents that may be combined to form a sinusoidal AC current. At a block **804**, a specified maximum current is compared with the plurality of output currents to generate an overcurrent signal. The overcurrent signal presents a fault signal responsive to one of the plurality of output currents $i_{\text{sub.a}}$ **112**, $i_{\text{sub.b}}$ **114**, and/or $i_{\text{sub.c}}$ **116**, flowing through at least one of the transistors **138**, **140**, **142**, **144**, **146**, and/or **148** and detectable at one of the output nodes **126**, **128**, and **130**, exceeding the specified maximum current **170**. At a block **806**, upon the fault signal being presented, one or more of the gate control signals **166** is replaced with a gate disable signal. At a block **808**, the gate disable signal is applied to the at least one of the transistors **138**, **140**, **142**, **144**, **146**, and/or **148** to prevent the output current $i_{\text{sub.a}}$ **112**, $i_{\text{sub.b}}$ **114**, and/or $i_{\text{sub.c}}$ **116** from flowing through the at least one of the transistors **138**, **140**, **142**, **144**, **146**, and/or **148**.

(36) This document describes systems and techniques for a current-limiting control strategy for single-loop droop-controlled grid-forming inverters. These systems and techniques may be realized using one or more of the entities or components shown in FIGS. 1-5B and used as described with reference to the example method of FIG. 8. However, the figures illustrate only some of the many possible systems capable of employing the described techniques.

(37) Unless context dictates otherwise, use herein of the word “or” may be considered use of an “inclusive or,” or a term that permits inclusion or application of one or more items that are linked by the word “or” (e.g., a phrase “A or B” may be interpreted as permitting just “A,” as permitting just “B,” or as permitting both “A” and “B”). Also, as used herein, a phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. For instance, “at least one of a, b, or c” can cover a, b, c, a-b, a-c, b-c, and a-b-c, as well as any combination with multiples of the same element (e.g., a-a, a-a-a, a-a-b, a-a-c, a-b-b, a-c-c, b-b, b-b-b, b-b-c, c-c, and c-c-c, or any other ordering of a, b, and c). Further, items represented in the accompanying figures and terms discussed herein may be indicative of one or more items or terms, and thus reference may be made interchangeably to single or plural forms of the items and terms in this written description.

Additional Examples

(38) Some additional examples of systems and techniques for a current-limiting control strategy for single-loop droop-controlled grid-forming inverters are as follows:

(39) Example 1. A current-limiting control system for a single-loop droop controller comprising: a hysteresis module configured to compare an output current detected across one or more transistors in an inverter controlled by the single-loop droop converter with a specified maximum current and to generate an overcurrent signal, the overcurrent signal presenting a fault signal responsive to the output current exceeding the specified maximum current; and a logic array configured to logically combine gate control signals generated by the single-loop droop controller to selectively direct the one or more transistors to allow the output current to flow therethrough with the overcurrent signal to present modified gate control signals to the one or more transistors, the logic array being configured to replace one or more of the gate control signals in the modified gate control signals with a gate disable signal responsive to the overcurrent signal presenting the fault signal.

(40) Example 2. The current-limiting control system of Example 1, wherein the hysteresis module includes a plurality of comparators, each of the comparators being configured to compare one of a plurality of output currents detected at a plurality of output nodes of the inverter with the specified

maximum current and to present the fault signal responsive to the one of the plurality of output currents exceeding the specified maximum current.

(41) Example 3. The current-limiting control system of Example 2, wherein each of the plurality of comparators in the hysteresis module is coupled with a buffer configured to maintain an output of the comparator.

(42) Example 4. The current-limiting control system of Example 1, wherein the one or more transistors includes a high-side transistor in a first branch of the inverter coupled between a positive side of a direct current bus and a first output node and a low-side transistor in a second branch of the inverter and coupled between a second output node and a negative side of the direct current bus, and wherein the output current is detected between the first output node and the second output node.

(43) Example 5. The current-limiting control system of Example 4, wherein the logic array includes a plurality of logic elements configured to replace the gate control signals with the gate disable signal in the modified gate control signals presented to the high-side transistor and the low-side transistor to prevent the output current from flowing through the high-side transistor and low-side transistor when the fault signal is presented.

(44) Example 6. The current-limiting control system of Example 5, wherein the plurality of logic elements includes a plurality of AND gates configured to replace one or more of the gate control signals with the gate disable signal in the modified gate control signals presented to the high-side transistor and the low-side transistor when the fault signal is presented to prevent the output current from flowing through the high-side transistor and the low-side transistor.

(45) Example 7. The current-limiting control system of Example 4, wherein the inverter is configured to redirect the output current to the direct current bus responsive to the gate disable signal being applied to the one or more transistors.

(46) Example 8. The current-limiting control system of Example 7, further comprising a plurality of diodes, each of the diodes being coupled between an emitter and a collector of each of the one or more transistors wherein the output current is redirected through the diodes to the direct current bus.

(47) Example 9. The current-limiting control system of Example 1, wherein the one or more transistors in an inverter include insulated-gate bipolar transistors.

(48) Example 10. A grid-forming inverter comprising: an inverter including a plurality of transistors configured in a three-phase bridge; a single-loop droop controller configured to monitor a plurality of output currents from the inverter and to generate a plurality of gate control signals applicable to the plurality of transistors to selectively direct the plurality of transistors to allow the output current to flow therethrough; and a current-limiting control system including: a hysteresis module configured to compare an output current detected across one or more transistors in an inverter controlled by the single-loop droop converter with a specified maximum current and to generate an overcurrent signal, the overcurrent signal presenting a fault signal responsive to the output current exceeding the specified maximum current; and a logic array configured to logically combine the plurality of gate control signals with the overcurrent signal to present modified gate control signals to the one or more transistors, the logic array being configured to replace one or more of the gate control signals in the modified gate control signals with a gate disable signal responsive to the overcurrent signal presenting the fault signal to prevent the output current from flowing through the one or more transistors.

(49) Example 11. The grid-forming inverter of Example 10, wherein the hysteresis module includes a plurality of comparators, each of the comparators being configured to compare one of a plurality of output currents detected at a plurality of output nodes of the inverter with the specified maximum current and to present the fault signal responsive to the one of the plurality of output currents exceeding the specified maximum current.

(50) Example 12. The grid-forming inverter of Example 10, wherein the one or more transistors

includes a high-side transistor in a first branch of the inverter coupled between a positive side of a direct current bus and a first output node and a low-side transistor in a second branch of the inverter and coupled between a second output node and a negative side of the direct current bus, and wherein the output current is detected between the first output node and the second output node.

(51) Example 13. The grid-forming inverter of Example 12, wherein the logic array includes a plurality of logic elements configured to replace the gate control signals with the gate disable signal in the modified gate control signals presented to the high-side transistor and the low-side transistor to prevent the output current from flowing through the high-side transistor and low-side transistor when the fault signal is presented.

(52) Example 14. The grid-forming inverter of Example 13, wherein the plurality of logic elements includes a plurality of AND gates configured to replace one or more of the gate control signals with the gate disable signal in the modified gate control signals presented to the high-side transistor and the low-side transistor when the fault signal is presented to prevent the output current from flowing through the high-side transistor and the low-side transistor.

(53) Example 15. The grid-forming inverter of Example 12, wherein the inverter is configured, responsive to the gate disable signal being applied to the one or more transistors, to redirect the output current to the direct current bus.

(54) Example 16. The grid-forming inverter of Example 15, further comprising a plurality of diodes, each of the diodes being coupled between an emitter and a collector of each of the one or more transistors wherein the output current is redirected through the diodes to the direct current bus.

(55) Example 17. The grid-forming inverter of Example 10, wherein the one or more transistors in an inverter include insulated-gate bipolar transistors.

(56) Example 18. A method of limiting current across one or more transistors in a grid-forming inverter controlled by a single-loop droop controller comprising: generating a plurality of gate control signals applicable to a plurality of transistors in the grid-forming inverter to cause the grid-forming inverter to generate a plurality of output currents combinable to form an alternating current; comparing a specified maximum current with the plurality of output currents to generate an overcurrent signal, the overcurrent signal presenting a fault signal responsive to one of the plurality of output currents flowing through at least one of the transistors exceeding the specified maximum current; replacing one or more of the gate control signals with a gate disable signal responsive to the fault signal being presented; and applying the gate disable signal to the at least one of the transistors to prevent the output current from flowing through the at least one of the transistors.

(57) Example 19. The method of Example 18, further comprising logically combining the overcurrent signal with the one or more gate control signals at a plurality of AND gates configured to replace one or more of the gate control signals with the gate disable signal responsive to the fault signal being presented by the overcurrent signal.

(58) Example 20. The method of Example 18, further comprising preventing the output current from flowing through the at least one of the transistors by redirecting the output current to a direct current bus to which the plurality of transistors is coupled.

CONCLUSION

(59) Although implementations of systems and techniques for a current-limiting control strategy for single-loop droop-controlled grid-forming inverters have been described in language specific to certain features and/or methods, the subject of the appended claims is not necessarily limited to the specific features or methods described. Rather, the specific features and methods are disclosed as example implementations of systems and techniques for a current-limiting control strategy for single-loop droop-controlled grid-forming inverters.

Claims

1. A current-limiting control system for a single-loop droop controller comprising: a hysteresis module configured to compare an output current detected across one or more transistors in an inverter controlled by the single-loop droop converter with a specified maximum current and to generate an overcurrent signal, the overcurrent signal presenting a fault signal responsive to the output current exceeding the specified maximum current; and a logic array configured to logically combine gate control signals generated by the single-loop droop controller to selectively direct the one or more transistors to allow the output current to flow therethrough with the overcurrent signal to present modified gate control signals to the one or more transistors, the logic array being configured to replace one or more of the gate control signals in the modified gate control signals with a gate disable signal responsive to the overcurrent signal presenting the fault signal.
2. The current-limiting control system of claim 1, wherein the hysteresis module includes a plurality of comparators, each of the comparators being configured to compare one of a plurality of output currents detected at a plurality of output nodes of the inverter with the specified maximum current and to present the fault signal responsive to the one of the plurality of output currents exceeding the specified maximum current.
3. The current-limiting control system of claim 2, wherein each of the plurality of comparators in the hysteresis module is coupled with a buffer configured to maintain an output of the comparator.
4. The current-limiting control system of claim 1, wherein the one or more transistors includes a high-side transistor in a first branch of the inverter coupled between a positive side of a direct current bus and a first output node and a low-side transistor in a second branch of the inverter and coupled between a second output node and a negative side of the direct current bus, and wherein the output current is detected between the first output node and the second output node.
5. The current-limiting control system of claim 4, wherein the logic array includes a plurality of logic elements configured to replace the gate control signals with the gate disable signal in the modified gate control signals presented to the high-side transistor and the low-side transistor to prevent the output current from flowing through the high-side transistor and low-side transistor when the fault signal is presented.
6. The current-limiting control system of claim 5, wherein the plurality of logic elements includes a plurality of AND gates configured to replace one or more of the gate control signals with the gate disable signal in the modified gate control signals presented to the high-side transistor and the low-side transistor when the fault signal is presented to prevent the output current from flowing through the high-side transistor and the low-side transistor.
7. The current-limiting control system of claim 4, wherein the inverter is configured to redirect the output current to the direct current bus responsive to the gate disable signal being applied to the one or more transistors.
8. The current-limiting control system of claim 7, further comprising a plurality of diodes, each of the diodes being coupled between an emitter and a collector of each of the one or more transistors, wherein the output current is redirected through the diodes to the direct current bus.
9. The current-limiting control system of claim 1, wherein the one or more transistors in an inverter include insulated-gate bipolar transistors.
10. A grid-forming inverter comprising: an inverter including a plurality of transistors configured in a three-phase bridge; a single-loop droop controller configured to monitor a plurality of output currents from the inverter and to generate a plurality of gate control signals applicable to the plurality of transistors to selectively direct the plurality of transistors to allow the output current to flow therethrough; and a current-limiting control system including: a hysteresis module configured to compare an output current detected across one or more transistors in an inverter controlled by the single-loop droop converter with a specified maximum current and to generate an overcurrent signal, the overcurrent signal presenting a fault signal responsive to the output current exceeding the specified maximum current; and a logic array configured to logically combine the plurality of

gate control signals with the overcurrent signal to present modified gate control signals to the one or more transistors, the logic array being configured to replace one or more of the gate control signals in the modified gate control signals with a gate disable signal responsive to the overcurrent signal presenting the fault signal to prevent the output current from flowing through the one or more transistors.

11. The grid-forming inverter of claim 10, wherein the hysteresis module includes a plurality of comparators, each of the comparators being configured to compare one of a plurality of output currents detected at a plurality of output nodes of the inverter with the specified maximum current and to present the fault signal responsive to the one of the plurality of output currents exceeding the specified maximum current.

12. The grid-forming inverter of claim 10, wherein the one or more transistors includes a high-side transistor in a first branch of the inverter coupled between a positive side of a direct current bus and a first output node and a low-side transistor in a second branch of the inverter and coupled between a second output node and a negative side of the direct current bus, and wherein the output current is detected between the first output node and the second output node.

13. The grid-forming inverter of claim 12, wherein the logic array includes a plurality of logic elements configured to replace the gate control signals with the gate disable signal in the modified gate control signals presented to the high-side transistor and the low-side transistor to prevent the output current from flowing through the high-side transistor and low-side transistor when the fault signal is presented.

14. The grid-forming inverter of claim 13, wherein the plurality of logic elements includes a plurality of AND gates configured to replace one or more of the gate control signals with the gate disable signal in the modified gate control signals presented to the high-side transistor and the low-side transistor when the fault signal is presented to prevent the output current from flowing through the high-side transistor and the low-side transistor.

15. The grid-forming inverter of claim 12, wherein the inverter is configured, responsive to the gate disable signal being applied to the one or more transistors, to redirect the output current to the direct current bus.

16. The grid-forming inverter of claim 15, further comprising a plurality of diodes, each of the diodes being coupled between an emitter and a collector of each of the one or more transistors wherein the output current is redirected through the diodes to the direct current bus.

17. The grid-forming inverter of claim 10, wherein the one or more transistors in an inverter include insulated-gate bipolar transistors.

18. A method of limiting current across one or more transistors in a grid-forming inverter controlled by a single-loop droop controller comprising: generating a plurality of gate control signals applicable to a plurality of transistors in the grid-forming inverter to cause the grid-forming inverter to generate a plurality of output currents combinable to form an alternating current; comparing a specified maximum current with the plurality of output currents to generate an overcurrent signal, the overcurrent signal presenting a fault signal responsive to one of the plurality of output currents flowing through at least one of the transistors exceeding the specified maximum current; replacing one or more of the gate control signals with a gate disable signal responsive to the fault signal being presented; and applying the gate disable signal to the at least one of the transistors to prevent the output current from flowing through the at least one of the transistors.

19. The method of claim 18, further comprising logically combining the overcurrent signal with the one or more gate control signals at a plurality of AND gates configured to replace one or more of the gate control signals with the gate disable signal responsive to the fault signal being presented by the overcurrent signal.

20. The method of claim 18, further comprising preventing the output current from flowing through the at least one of the transistors by redirecting the output current to a direct current bus to which the plurality of transistors is coupled.

